

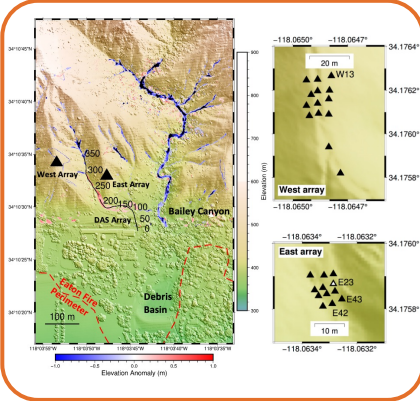
Using Small Aperture Nodal Arrays and Distributed Acoustic Sensing to Monitor Post-Eaton Fire Debris Flows

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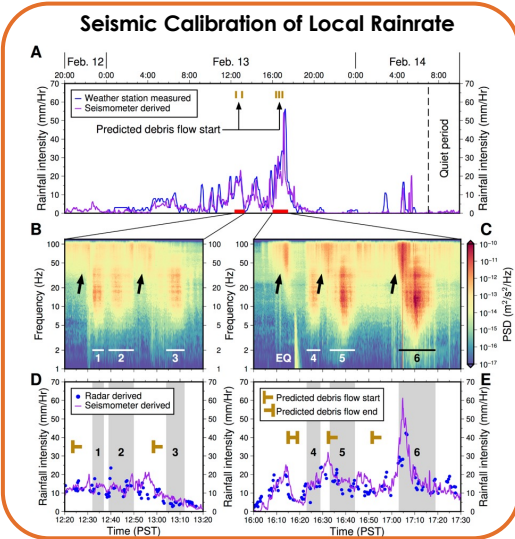


Post-Fire Debris Flows in San Gabriel Foothills

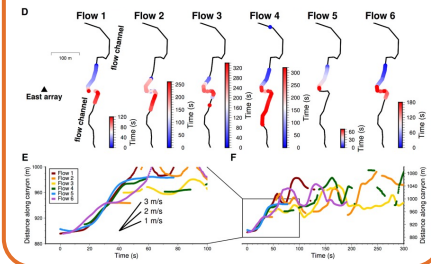


As the Bailey Canyon region in the foothills of the San Gabriels was notably burned in the Eaton Fire (LA), the question of debris flows in the canyon became **when rather than if**. On February 13, 2025, a high-intensity rainfall event in the San Gabriel foothills triggered regional debris flows. We rapidly deployed two small-aperture seismic arrays and a 1 km DAS array the day before the event, capturing key flow characteristics and providing insight into their mechanistic properties.

Seismic Determination of Precipitation and Flow Evolution



Beamforming Illuminates Flow Events



High-frequency (>60 Hz) signals closely tracked precipitation rates transforming the seismic sensors into high-resolution rain gauges. Lower frequency (10–40 Hz) signal correlated with camera observations of debris basin activity, originating from debris flow events. Events show a migrating flow front traveling approximately 370 meters in 100 seconds, yielding a flow speed of ~3.7 m/s.

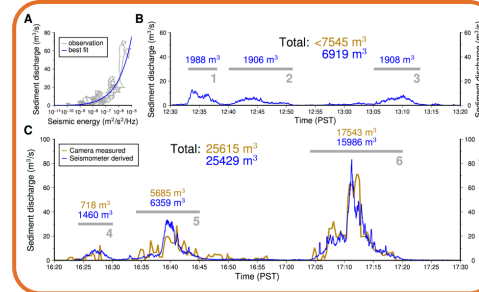
References

Wei, X., E. Geyman, Z. Chen, A. Reid-McLaughlin, M. P. Lamb, and Z. Zhan (in prep.), Resolving the triggering and evolution of debris flows following the Eaton Fire with seismometers.
Reid-McLaughlin, A., Z. Chen, X. Wei, E. Biondi, and Z. Zhan (in prep.), Rapid response DAS deployment for in-situ monitoring of post-Eaton Fire debris flows.
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Conclusions

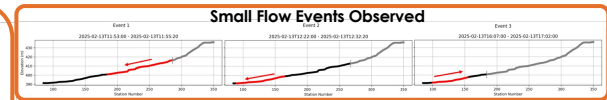
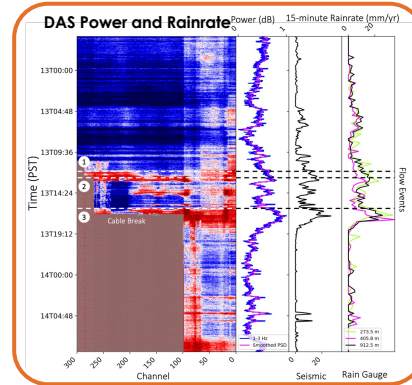
Seismic results highlight the **effectiveness of small-aperture seismic arrays** for monitoring both **precipitation** and **debris-flow dynamics**. The study demonstrates the **feasibility of rapid-response DAS** as a powerful tool for **post-fire debris-flow monitoring** in the **San Gabriel Mountains**. Arrays installed **less than 24 hours before** the forecasted rainfall events and **successfully captured** both the **evolution of rainfall-related seismic noise**, and the **onset and propagation of distinct debris-flow events**.

Seismically Estimated Debris Discharge



For debris flows 4-6, where we have both the seismic and camera time series data, we perform a cross-correlation to align the two records in time and then fit their relationship in the semi-logarithmic space with the minimum least-squares misfit. Using this fitting procedure, the **observed debris flow volume of 25,521 m³ matches very well with the volume derived from seismic energy, which is 25,482 m³**. Using this relationship, we derive total volume for flows 1-3 to be 6,923 m³, agreeing with the observed value of 7,686 m³ at the beginning of the camera recording.

DAS In-Situ Flow Monitoring of side catchment



The **coherence between DAS power and rainfall intensity** at reference channels across the entire day allows us to observe spatial evolution of DAS power across the cable. At the reference site (channel 50), DAS power increases with rising rain intensity, peaks at ~14.4 mm/hr, and then decreases as rainfall subsides. In the catchment (channel 100 - 300), the sustained power until just after 12:00 PST indicates that the signal became dominated by the debris flow itself. Given the high spatial and temporal sampling of DAS, we **can observe the meter scale propagation of debris flow events by tracking the front of our high strainrate signal across time**.

